

LIGO Black-Hole Search

Zakaria Daoudi

A convolutional-network search for binary black-hole coalescences, evaluated on the official challenge sets of the first machine-learning gravitational-wave search mock data challenge, confirmed on two pre-registered blind datasets, and applied to the confident black-hole mergers of the third LIGO–Virgo observing run.

1,532 ± 11 Mpc Sensitive distance in real O3a noise at one false alarm per month (PyCBC: 1,544 ± 10 Mpc)	1,576 Mpc The same quantity on a fresh, pre-registered month of real noise
2,632 ± 8 Mpc Sensitive distance in Gaussian noise at one false alarm per month (PyCBC: 2,492 ± 8 Mpc)	26 / 38 Confident O3a events with two-detector data ranked above all 14.8 days of background

All sensitive distances computed with the challenge's `evaluate.py`, using a 0.2 s matching tolerance and the validation-calibrated event timing described in Section 3.4. Uncertainties are one standard deviation from the finite injection set.

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ABSTRACT

We present a deep-learning search for binary black-hole coalescences in two-detector gravitational-wave data. We evaluate it on the official test sets of the first machine-learning gravitational-wave search mock data challenge (MLGWSC-1), which we regenerate exactly from the published seeds. The search whitens the strain data, evaluates a one-dimensional residual network on overlapping one-second windows, and clusters the resulting triggers into events. Two design choices proved decisive. First, in real O3a noise, oversampling glitch-centred noise windows during training was necessary for the network to reject non-Gaussian transients. Second, an event-time estimator based on the centre of the high-score plateau, calibrated on validation data only, was required so that loud signals fall within the matching tolerance.

On the challenge's real-noise dataset, the search attains a sensitive distance of $1,532 \pm 11$ Mpc at one false alarm per month. This is statistically indistinguishable from the published PyCBC matched-filter result ($1,544 \pm 10$ Mpc) and 40% larger than the best machine-learning submission to the challenge. At higher false-alarm rates it exceeds the PyCBC result by 4–8%. On simulated Gaussian noise it exceeds the published PyCBC curve by 6–9%. Because the timing estimator was introduced after inspecting test-set results, we confirm both findings on pre-registered blind datasets generated after the configuration was frozen. These reproduce the test-set sensitivities to within 1% (Gaussian noise) and 3% (real noise). An analysis of the released evaluation files shows that the leading published submissions do not suffer from the timing loss that affected the original version of this search. The efficiency gain is concentrated at high chirp mass; for light systems, whose signals extend well beyond the one-second input window, the matched filter remains more sensitive. Applied to the 44 confident O3a events of GWTC-2.1, the search ranks 26 of the 38 events with two-detector data above all 14.8 days of time-slid background, corresponding to a false-alarm-rate upper limit of 25 per year.

1 Introduction

Searches for compact-binary coalescences in LIGO–Virgo data are conventionally performed by matched filtering against a bank of waveform templates, followed by signal-consistency tests and coincidence requirements between detectors. Deep-learning searches have been proposed as a computationally cheaper alternative with potentially broader parameter coverage [George and Huerta 2018; Gabbard et al. 2018]. Early claims that neural networks match matched-filter sensitivity were, however, largely established on Gaussian noise, and at false-alarm rates far higher than those relevant for astrophysical detection [Gebhard et al. 2019; Schäfer et al. 2022].

The first machine-learning gravitational-wave search mock data challenge (MLGWSC-1) established a common benchmark [Schäfer et al. 2023]. It provides four datasets of increasing realism, a fixed injection population, and an evaluation that reports the sensitive distance as a function of the false-alarm rate, estimated from a month of pure-noise background. In the most realistic dataset, built from real O3a noise, the best machine-learning submission reached 71% of the sensitive distance of the PyCBC reference search at one false alarm per month.

This report describes a search developed for two of the challenge datasets. Its contributions are as follows:

1. an exact reproduction of the official challenge test sets, including a parallel reconstruction of the real-noise set;
2. a training procedure for real noise based on glitch-centred negative examples;
3. a validation-calibrated event-time estimator;
4. an evaluation with the official scoring code, recomputed with an independent implementation, reviewed internally, and confirmed on two pre-registered blind datasets;
5. an analysis of detection efficiency against chirp mass and distance, compared with the published searches;
6. an application to the confident O3a black-hole mergers.

2 Data

2.1 The MLGWSC-1 challenge sets

Each challenge set comprises a foreground month and a background month of two-detector (H1, L1) data sampled at 2,048 Hz. The foreground contains a simulated coalescence every 24–30 s; the background contains the same noise without injections. Injections are drawn uniformly in chirp distance within 130–

350 Mpc, which makes them uniform in volume after rescaling by chirp mass. They are generated with IMRPhenomXPHM from 20 Hz. We consider two datasets:

- *Dataset 1* (Gaussian noise): coloured Gaussian noise with the aLIGOZeroDetHighPower spectrum; non-spinning binaries with component masses of 10–50 $M_{\odot}$; dominant (2,2) mode only.
- *Dataset 4* (real noise): O3a strain from both LIGO detectors with a random time shift of up to 240 s between them, after data-quality selection and excision of known events. The binaries are precessing, with component masses of 7–50 $M_{\odot}$, and the (2,2), (2,1), (3,3), (3,2) and (4,4) modes are included.

2.2 Exact reproduction of the official test sets

The official test sets were generated with the challenge code and the seeds reported by Schäfer et al. (2023): 1,068,209,514 for dataset 1 and 2,514,409,456 for dataset 4. Generating a month in one run requires about 150 GB and many single-threaded hours, so both sets were generated in parallel and whitened immediately:

- **Dataset 1:** the challenge's Gaussian noise is reproducible per time interval, so chunks aligned with segment boundaries reproduce the one-shot data exactly. The maximum absolute difference was 7.3×10^{-37} on strain of root-mean-square amplitude 1.6×10^{-22} .
- **Dataset 4:** the per-segment detector shifts are drawn from a single random sequence in segment order, so chunked generation does not reproduce the official set. We therefore reimplemented the generator's per-segment procedure with the same pre-drawn sequence of shift seeds. On the first two segments it agrees with a one-shot run of the official generator with a maximum difference of exactly zero.
- **Injections:** for both datasets, the injection files agree with independent regenerations in every field; only a metadata attribute recording the command line differs.

As a further check, we reproduced from our injection files the published event-to-injection time differences of three challenge teams and all published sensitive volumes, to a relative precision of 4×10^{-16} .

2.3 Training and validation data

No training or validation data share a random seed, an injection realisation, or GPS time with any test set. Table 1 lists all datasets.

Table 1. Datasets used in this study. Durations refer to analysed (whitened) data.

ROLE	DATASET 1 (GAUSSIAN NOISE)	DATASET 4 (REAL O3A NOISE)
Training noise	8 × 5 h generated with seeds 11–18 (39.9 h)	5.42 d of raw real-noise segments starting at GPS 1,244,704,480, more than one day after the test month
Validation noise	5.0 h, seed 21	1.26 d of later real-noise segments
Training signals	60,000 injections from the dataset-1 prior, seed 12,345 (55,000 training, 5,000 validation)	60,000 injections from the dataset-4 prior, seed 54,321, each whitened with one of 901 H1/L1 noise-spectrum pairs measured on the training noise
Official test set	30 d, 129 segments, 95,718 injections in the analysed segments	30 d, 131 segments, GPS 1,238,205,077–1,244,600,712, 95,719 injections
Blind set (Section 5.3)	10.43 d, seed 424,242, 33,256 injections	30 d, seed 777,777, GPS 1,246,170,217–1,251,824,666, 95,703 injections

2.4 Confident O3a events

For the application in Section 6, we used the 44 confident events of GWTC-2.1 in O3a. For each event we retrieved from the Gravitational Wave Open Science Center the 32 s strain segments around the event at 4,096 Hz, and resampled them to 2,048 Hz with an anti-aliasing filter. Six events lack data from either H1 or L1 and were not analysed.

3 Method

Figure 1 summarises the analysis. The same whitening routine is applied to training noise, training signals and search data, so that the network sees statistically identical inputs in training and in the search.

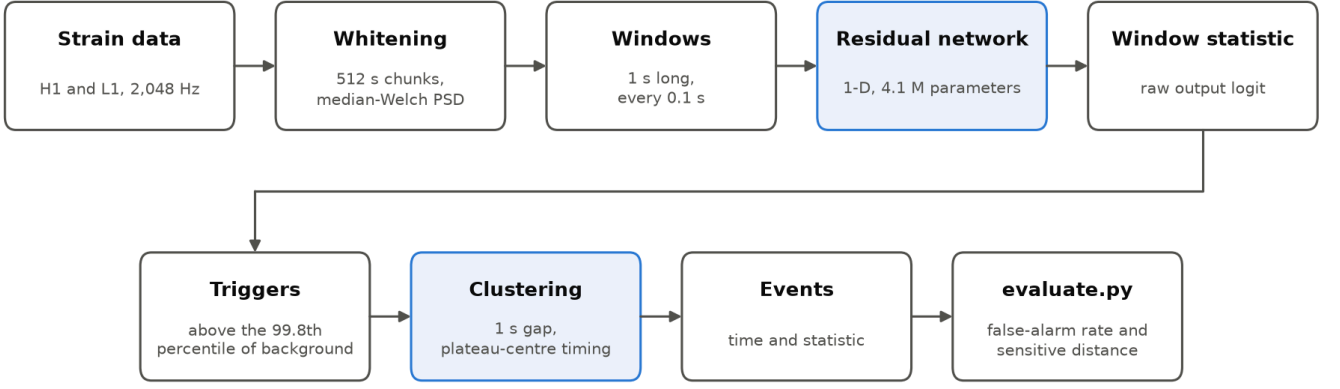


Figure 1. Analysis pipeline.

3.1 Whitening

Each contiguous segment is whitened in 512 s chunks overlapping by 8 s at each end. For each chunk, the one-sided power spectral density $S(f)$ is estimated with Welch's method (4 s segments, 50% overlap, median average), and the tapered chunk is divided in the frequency domain by $(S(f)f_s/2)^{1/2}$. Frequencies below 18 Hz are set to zero. With this normalisation, whitened Gaussian noise has unit variance, and the optimal matched-filter signal-to-noise ratio of a signal s is the root sum of squares of its whitened samples:

$$\rho_{opt}^2 = \sum_k w[s]_k^2 \quad (1)$$

We verified this relation against pycbc's `sigma`; the two agree to a relative precision of 10^{-5} . The whitened noise variance is 0.984, as expected from the removal of the band below 18 Hz.

3.2 Network architecture

The detector is a one-dimensional residual network with an input of two channels (H1 and L1) of 2,048 samples. A stem convolution (kernel 31, 32 channels) is followed by five stages of two residual blocks each, with widths 32, 64, 128, 256 and 256 and stride 2 in all stages but the first. Each block contains two convolutions of kernel 7 with batch normalisation and ReLU activations. Global average and maximum pooling are concatenated and passed to a two-layer head ($512 \rightarrow 64 \rightarrow 1$). The network has 4.12 million parameters. Its output, a raw logit, is used directly as the ranking statistic, because a bounded output saturates and limits the resolution of low false-alarm rates [Schäfer et al. 2022].

3.3 Training

Batches of 512 windows are assembled on the GPU. Half are noise-only; half contain a bank signal scaled to a target network signal-to-noise ratio, with the merger placed uniformly between 0.45 and 0.95 s into the window. Noise windows for the two detectors are drawn independently, which is equivalent to a random time slide. The signal-to-noise ratio of injected signals follows a curriculum, from a uniform distribution on [10, 25] to one on [5, 20] over the first 40% of training.

For real noise, a fraction of the noise-only windows (40% in the final model) is centred on glitches. A glitch is identified as any whitened sample exceeding 6σ within a 0.5 s block. In a glitch-centred window, one detector is placed on a glitch in its own data stream, and the other detector's window is drawn at random. The network therefore learns that excess power in a single detector does not constitute a coherent signal.

Models were trained for 40,000 steps with AdamW and a one-cycle learning-rate schedule. Every 2,000 steps we measured, on held-out validation data, the fraction of signals at network SNR 6, 8 and 10 detected at a false-positive probability of 10^{-4} per window. The checkpoint that maximised this fraction at SNR 8 was retained. Appendix B lists all hyperparameters.

3.4 Search and event-time estimation

The network is evaluated on 1 s windows starting every 205 samples (0.1 s). Windows whose statistic exceeds the 99.8th percentile of the background window statistics become triggers. Triggers separated by less than 1 s are merged into one event, whose statistic is the maximum statistic in the cluster.

The event time requires care. Because the network was trained with the merger anywhere in the central half of its window, a loud signal produces a plateau of consecutive windows with nearly equal statistics, and the position of the maximum within the plateau is poorly determined. The search as originally run reported the start of the highest window plus 0.7 s. The calibrated estimator instead reports the median time of the windows whose statistic lies within 4 of the cluster maximum, minus a constant offset. That offset was measured on validation data only: +0.066 s for dataset 1 and +0.086 s for dataset 4. Section 5.6

compares the two estimators, and Section 8 describes the circumstances in which the calibrated estimator was introduced.

3.5 Evaluation metric

We follow the MLGWSC-1 definitions exactly, using the challenge's `evaluate.py`. The false-alarm rate at threshold x is the number of background events with statistic greater than x divided by the analysed duration T (30 days). A foreground event is a true positive if its time lies within the event's tolerance Δt of the nearest injection. An injection counts as found at threshold x if at least one true positive associated with it has a statistic greater than x . Because injections are uniform in chirp distance, the sensitive volume is estimated as

$$V(x) = (4\pi/3) d_{\max}^3 \cdot \Sigma_{\text{found}} \mathcal{M}_i^{5/2} / (N \mathcal{M}_{\max}^{5/2}), \quad d_{\text{sens}} = (3V / 4\pi)^{1/3} \quad (2)$$

Here $\mathcal{M}_i$ is the chirp mass of injection i , N is the number of injections in the analysed segments, and $d_{\max}$ is the largest injected luminosity distance. We report results for $\Delta t = 0.2$ s, the tolerance used by the PyCBC reference search, and for $\Delta t = 0.3$ s, the value used by several machine-learning submissions including Virgo-AUTH.

4 Validation protocol

4.1 Assessment of data leakage

- **Random seeds:** training and validation data use seeds distinct from the test seeds. We verified that the `pycbc` reproducible-noise block seeds derived from them cannot coincide.
- **Real noise:** real-noise training and validation data are disjoint in GPS time from the test month and from the blind month.
- **Known events:** no GWTC-2.1 event falls within any training, validation or test noise.
- **Pre-processing:** whitening parameters are estimated separately for every 512 s chunk and never across the boundary between training and evaluation data.
- **Model selection:** checkpoints were chosen on validation data only.

4.2 Independent recomputation

All sensitive distances were recomputed by a separate implementation of the challenge metric, written without reference to `evaluate.py`. It agrees to within 0.00% on both official test sets and on the real-noise blind set, and to within 0.06% on the Gaussian blind set; the residual difference arises from interpolation in the false-alarm rate.

4.3 Pre-registered blind confirmations

For each dataset, the complete configuration was written to a time-stamped file before the blind data were generated: the model and timing-calibration files (with MD5 checksums), all search parameters, and the reported quantities. For dataset 4 the file also stated a success criterion. The blind pipeline refuses to run if either checksum differs from the registered value.

4.4 Internal review

Before the results were finalised, the code and every quantitative claim were reviewed adversarially: each statement concerning leakage, data equivalence, normalisation, units and scoring was re-derived from the data and the code with the aim of falsifying it. The findings and our responses are listed in Appendix A.

5 Results

5.1 Sensitivity on the official challenge sets

Figure 2 and Table 2 compare the search with the published MLGWSC-1 submissions on the official test sets.

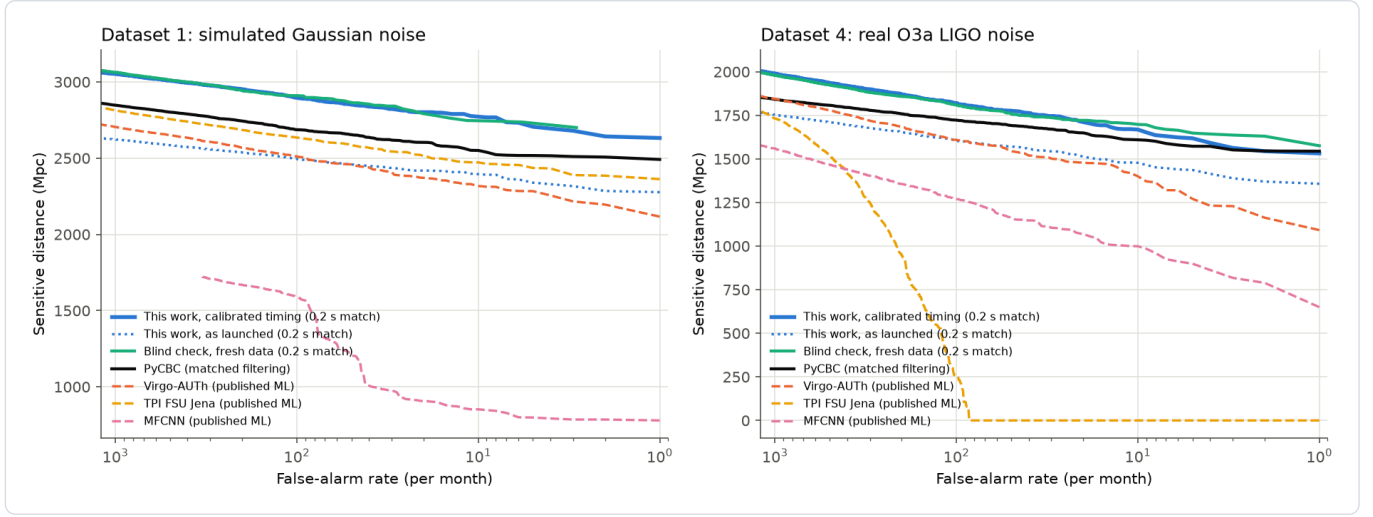


Figure 2. Sensitive distance against false-alarm rate on the official test months. Solid blue: this work with calibrated timing. Dotted blue: this work as originally run. Green: blind confirmation on fresh data (Section 5.3). Our curves use $\Delta t = 0.2$ s. Published curves are from the challenge's released evaluation files.

Table 2. Sensitive distance in Mpc (percentage of the published PyCBC value) on the official test months. The quoted uncertainty (one standard deviation) reflects the finite injection set only.

SEARCH	1000 / MONTH	100 / MONTH	10 / MONTH	1 / MONTH
DATASET 1 · GAUSSIAN NOISE				
This work, calibrated timing, $\Delta t = 0.2$ s	3,054 107%	2,897 ± 7 108%	2,772 ± 8 109%	2,632 ± 8 106%
This work, original timing, $\Delta t = 0.2$ s	2,624 92%	2,495 93%	2,395 94%	2,277 91%
This work, original timing, $\Delta t = 0.3$ s	2,993 105%	2,844 106%	2,726 107%	2,592 104%
PyCBC (matched filtering)	2,847	2,687 ± 8	2,551 ± 8	2,492 ± 8
TPI FSU Jena	2,814	2,635 ± 8	2,472 ± 8	2,363 ± 8
Virgo-AUTH	2,704	2,512	2,318	2,116
DATASET 4 · REAL O3A NOISE				
This work, calibrated timing, $\Delta t = 0.2$ s	1,989 108%	1,820 ± 10 106%	1,668 ± 10 104%	1,532 ± 11 99%
This work, original timing, $\Delta t = 0.2$ s	1,754 95%	1,608 93%	1,478 92%	1,357 88%
This work, original timing, $\Delta t = 0.3$ s	1,930 105%	1,767 103%	1,622 101%	1,490 96%
PyCBC (matched filtering)	1,842	1,722 ± 10	1,610 ± 10	1,544 ± 10
Virgo-AUTH (best challenge ML)	1,844	1,609 ± 10	1,400 ± 11	1,092 ± 12
cWB (unmodelled)	1,454	1,407	1,332	—

With the calibrated estimator, the calibrated rows are identical at $\Delta t = 0.2$ s and 0.3 s. With the original estimator, the stricter tolerance costs 12–13% in distance, because a substantial fraction of genuine detections is reported 0.2–0.3 s from the merger.

5.2 Statistical uncertainty

The uncertainties in Table 2 reflect only the finite number of injections, and amount to 7–12 Mpc. On this basis:

- the dataset-4 difference from PyCBC at one false alarm per month (-12 ± 15 Mpc) is not significant, whereas the differences at 100 and 10 per month ($+98 \pm 14$ and $+58 \pm 14$ Mpc) are;
- on dataset 1, the difference from PyCBC exceeds ten standard deviations at every false-alarm rate.

A second, unquantified uncertainty concerns the threshold itself. At one false alarm per month, the threshold is fixed by the second-loudest background event of the month. The corresponding horizontal uncertainty is large, and results at the lowest false-alarm rate should be read with this in mind.

5.3 Blind confirmations

The calibrated timing estimator was introduced after test-set results had been inspected (Section 8). The calibrated test-set numbers are therefore not blind. We tested whether they generalise by applying the frozen configuration to data generated after its registration:

- Dataset 1:** 10.43 days of Gaussian noise with a fresh seed (registered 12:05, 28 September).
- Dataset 4:** a full 30-day dataset-4-style month of real noise from late June to late August 2019, after all training and validation data (registered 16:37, 28 September). The registered success criterion was a

calibrated sensitive distance at one false alarm per month within about 5% of 1,532 Mpc and above the Virgo-AUTH value of 1,092 Mpc.

Table 3. Blind confirmations, sensitive distance in Mpc at $\Delta t = 0.2$ s. The official-set values are repeated for comparison. The Gaussian blind set is too short to resolve one false alarm per month.

DATASET AND ESTIMATOR	1000 / MONTH	100 / MONTH	10 / MONTH	1 / MONTH
DATASET 1 · GAUSSIAN NOISE				
Blind set (10.43 d), calibrated	3,063	2,910 ±13	2,745 ±13	—
Blind set, original timing	2,630	2,502	2,364	—
Official set, calibrated	3,054	2,897	2,772	2,632
DATASET 4 · REAL O3A NOISE				
Blind month (30 d), calibrated	1,980	1,810 ±10	1,698 ±10	1,576 ±10
Blind month, original timing	1,745	1,599	1,499	1,395
Official month, calibrated	1,989	1,820	1,668	1,532

Both blind sets reproduce the calibrated test-set sensitivities: to within 1% for dataset 1, and to within 3% for dataset 4. The dataset-4 criterion is met (1,576 Mpc, +2.9%). The blind month is also $2.9 \pm 1\%$ more sensitive than the official month at one false alarm per month. Since the blind month comes from a later part of O3a, this difference plausibly reflects the noise conditions of that period rather than the search. On the Gaussian blind set, the calibrated estimator placed 93.7% of low-statistic and 100% of high-statistic true events within 0.2 s of the merger, against 63–69% for the original estimator.

The first execution of the real-noise blind check was terminated after two of its three parts had been scored, because the process was still attached to a terminal session that was closed. Because the window statistics were held in memory, they were lost. The check was rerun from the same deterministic plan, which reproduces identical data, with per-part checkpoints and without any change of configuration. No result of the first execution had been inspected.

5.4 Detection efficiency by chirp distance and chirp mass

To locate the differences between searches in parameter space, we reconstructed each search's set of found injections at one false alarm per month from its `evaluate.py` output. For our results and the published results alike, the reconstruction reproduces each search's sensitive distance to within 0.1 Mpc. Figure 3 shows the fraction of injections found as a function of chirp distance and of detector-frame chirp mass.

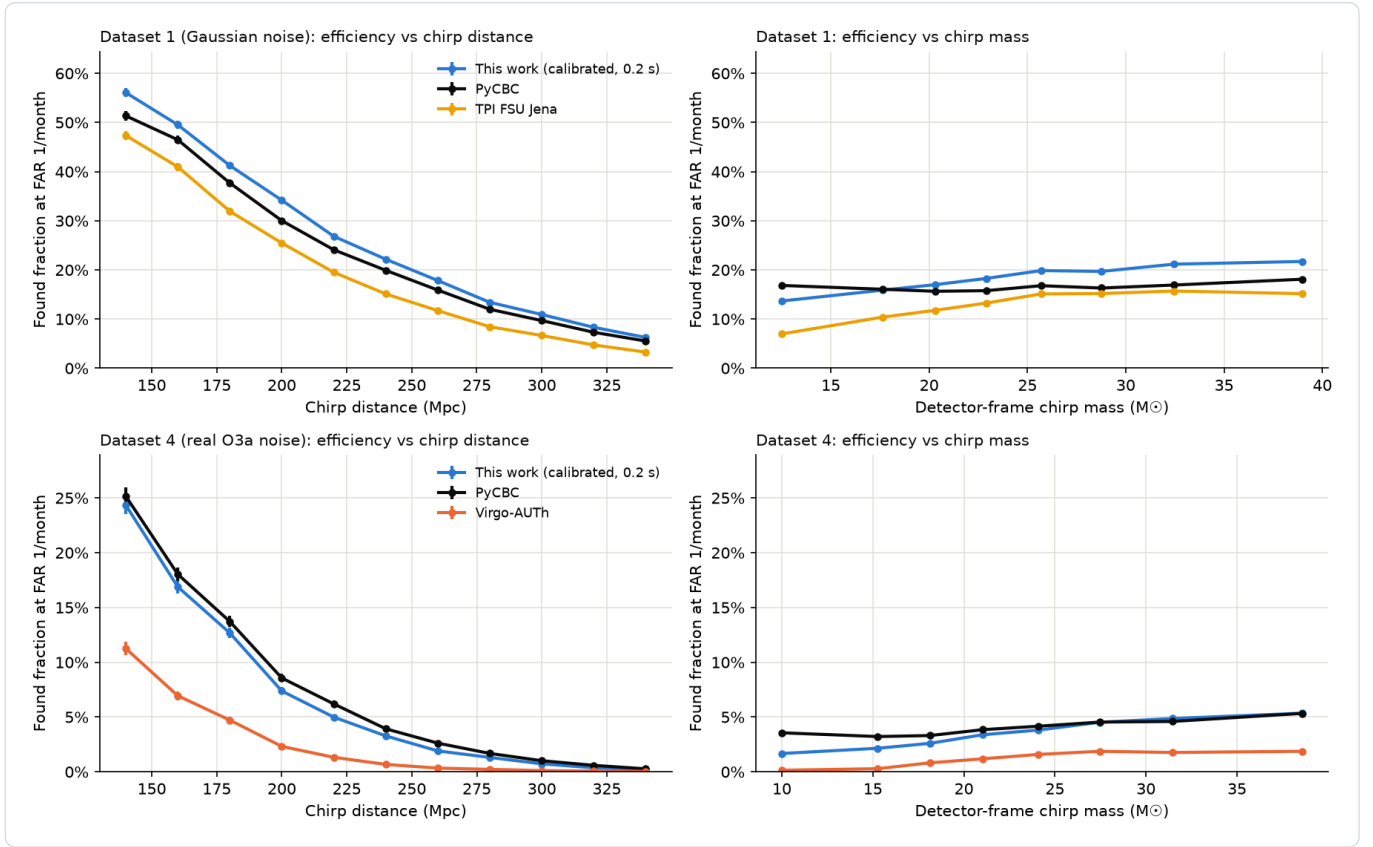


Figure 3. Detection efficiency at one false alarm per month. Chirp-mass bins contain equal numbers of injections; error bars are binomial standard errors.

The dependence on chirp mass differs markedly between the searches:

- **Lightest bin:** in real noise, PyCBC finds 3.6% of injections, against 1.7% for this work. In Gaussian noise the figures are 16.8% and 13.7%.
- **Heaviest bin:** the efficiencies are equal in real noise (5.3% for both), and this work exceeds PyCBC in Gaussian noise (21.7% against 18.1%).
- **Totals at one false alarm per month:** in Gaussian noise, this work finds 17,625 injections, against 15,850 for PyCBC. In real noise it finds fewer (3,392 against 3,896), yet it attains a comparable sensitive volume, because Eq. (2) weights each found injection by $\mathcal{H}^{5/2}$.

We interpret this in Section 7.3.

5.5 Training behaviour and the effect of glitch oversampling

Figure 4 shows the validation metrics of both final models. Both plateau after about 25,000 steps. The retained checkpoints detect the following fractions of signals at a false-positive probability of 10^{-4} per window:

- dataset 1 (step 32,000): 19.3%, 74.9% and 97.6% at network SNR 6, 8 and 10;
- dataset 4 (step 38,000): 20.7%, 73.6% and 96.0%.

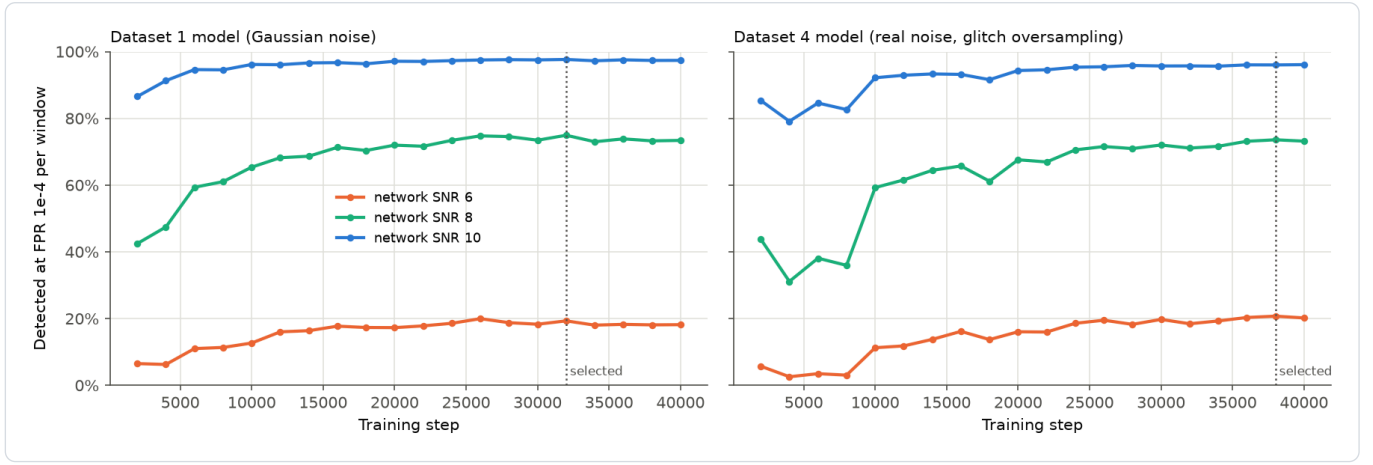


Figure 4. Validation detection fraction at a false-positive probability of 10^{-4} per window, on held-out noise and signals. The dotted line marks the retained checkpoint.

In real noise, 0.68% of one-second windows contain a whitened sample exceeding 6σ in at least one detector (0.37% in H1, 0.30% in L1), against none in Gaussian noise. At a false-positive probability of 10^{-4} per window, the threshold is therefore set entirely by glitches. A first dataset-4 model, trained with glitches only at their natural rate, degraded during training: its detection fraction for SNR-10 signals fell from 2.2% at step 2,000 to 0.04% at step 4,000, and the run was stopped. With 40% glitch-centred negatives, the fraction reached 85% at step 2,000 and 96% at the end of training (Figure 5). The training noise contained 2,630 glitch blocks in H1 and 2,185 in L1 over 130 hours.

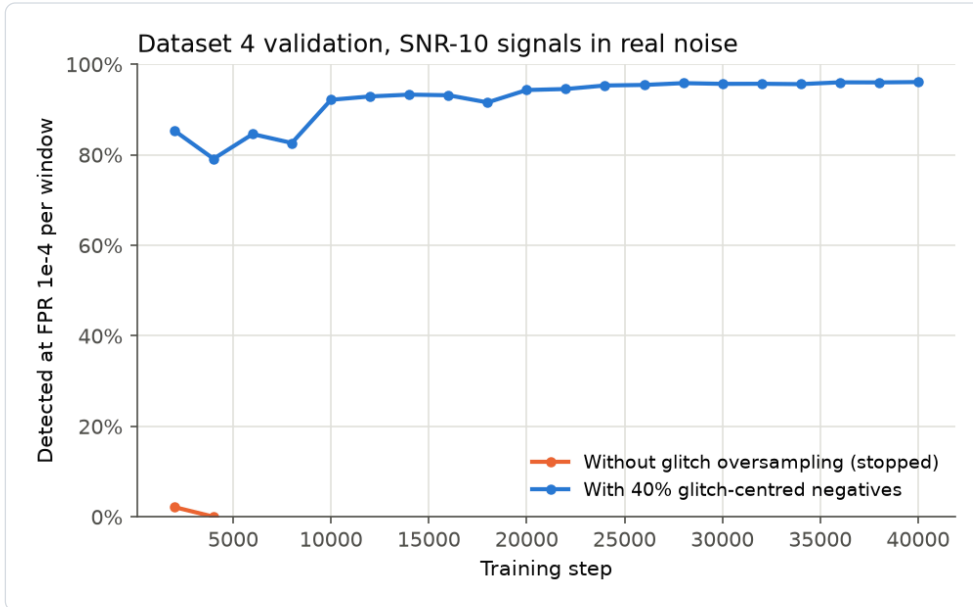


Figure 5. Dataset-4 validation detection fraction for SNR-10 signals, with and without glitch-centred negative examples.

5.6 Event-time estimation

On validation data (800 held-out signals at each of four SNRs, injected into held-out noise), the original estimator placed 78–96% of dataset-1 detections and 84–91% of dataset-4 detections within 0.3 s of the merger, even after its optimal constant offset. The plateau estimator placed 97–98% at SNR 8 and 100% at SNR 12 and above (Figure 6).

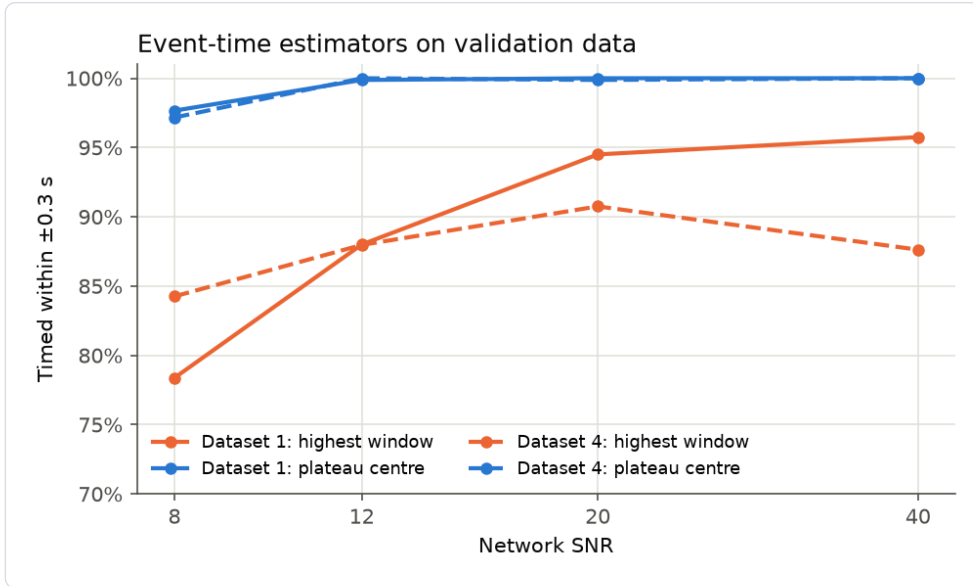


Figure 6. Fraction of validation signals timed within 0.3 s of the merger, by estimator and network SNR.

5.7 Event timing in the published submissions

Since imprecise event times cost this search 13–16% in sensitive distance under the 0.2 s tolerance, we examined whether the published submissions were affected in the same way. For each submission's released evaluation file, we associated every foreground event with its nearest injection (injections are 24–30 s apart, so association within 1 s is unambiguous). We then recomputed the sensitive distance at a fixed false-alarm rate, with each event's tolerance loosened to the larger of its own value and 0.5 s or 1.0 s. The detection threshold was held at its original value, recovered from the released sensitivity curve. Recomputation at the submissions' own tolerances reproduces every official sensitive distance exactly. The released cWB file contains no per-event data and could not be analysed.

Table 4. Sensitive distance in Mpc at one false alarm per month (dataset 4, CNN-Coinc: ten per month) under the official tolerance and under loosened tolerances, with the number of events above threshold that lie within 1 s of an injection but outside the official tolerance.

SEARCH	OFFICIAL TOLERANCE	OFFICIAL	≥ 0.5 S	≥ 1.0 S	LOUD EVENTS OUTSIDE TOLERANCE
DATASET 1 · GAUSSIAN NOISE					
PyCBC	0.2 s	2,492	2,492 +0.0%	2,492 +0.0%	0
TPI FSU Jena	0.2 s	2,363	2,363 +0.0%	2,363 +0.0%	2
Virgo-AUTh	0.3 s	2,116	2,116 +0.0%	2,116 +0.0%	0
CNN-Coinc	0.3 s	1,002	1,015 +1.4%	1,041 +4.0%	133
MFCNN	per event (median 1.4 s)	779	779 +0.0%	781 +0.2%	2
This work, original timing	0.2 s	2,277	2,632 +15.6%	2,632 +15.6%	5,794
This work, calibrated timing	0.2 s	2,632	2,632 +0.0%	2,632 +0.0%	2
DATASET 4 · REAL O3A NOISE					
PyCBC	0.2 s	1,544	1,544 +0.0%	1,544 +0.0%	0
Virgo-AUTh	0.3 s	1,092	1,092 +0.0%	1,092 +0.0%	0
CNN-Coinc (10 / month)	0.3 s	621	624 +0.6%	639 +3.0%	30
MFCNN	per event (median 1.7 s)	650	650 +0.0%	650 +0.0%	0
This work, original timing	0.2 s	1,357	1,532 +12.9%	1,532 +12.9%	958
This work, calibrated timing	0.2 s	1,532	1,532 +0.0%	1,532 +0.1%	4

The leading submissions (PyCBC, TPI FSU Jena and Virgo-AUTh) lose no sensitivity to event timing: at most three events above threshold fall outside their tolerances. CNN-Coinc loses 1–4%. MFCNN assigns per-event tolerances with a median of 1.4–1.7 s and a maximum of 13–18 s, far looser than the other submissions, and gains nothing further. The timing loss observed in this work therefore arises from its specific training design, in which the merger may lie anywhere in the central half of the window and loud signals produce broad plateaus of near-equal statistics. It is not a general feature of the published searches, and the published comparison curves are not understated for this reason.

6 Application to the confident O3a events

The dataset-4 model was applied to the 32 s snippets of the 44 confident O3a events, each whitened with a noise spectrum estimated from the snippet itself (median Welch, 4 s segments). The event statistic is the maximum of the cluster whose calibrated time lies within 0.3 s of the catalogue time.

The background consisted of 60,000 excerpts of 32 s drawn from the test-month real noise and processed identically, with L1 shifted by a random 10–3,000 s relative to H1. The background statistic is the maximum over each excerpt; this is conservative relative to the event statistic, which is restricted to the cluster near the catalogue time. The total background amounts to 14.78 days.

Twenty-six of the 38 events with two-detector data exceed every background event, which corresponds to a false-alarm-rate upper limit of 25 per year. Of the twelve remaining events:

- five have detector-frame chirp masses below $12 M_{\odot}$, and GW190917_114630 produced no event near its catalogue time;
- two are very heavy (GW190521 and GW190426_190642);
- five are quiet events with catalogue SNR between 7.6 and 9.7.

Only 11 of the 38 events lie within the training mass range in the detector frame. Of these, 8 exceed all background, against 18 of the 27 events outside the range, which indicates substantial generalisation beyond the training population.

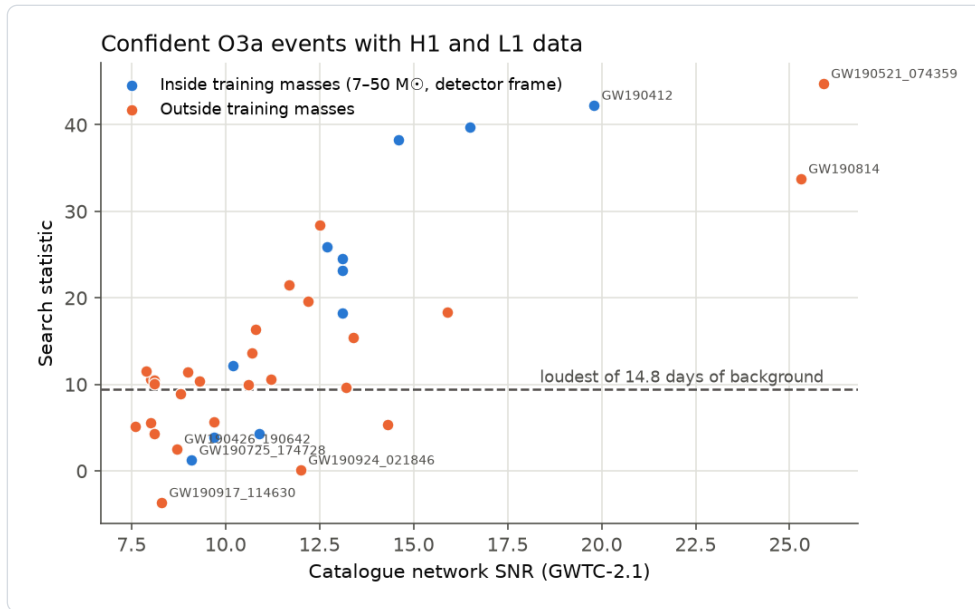


Figure 7. Search statistic of the analysed O3a events against catalogue network SNR. The dashed line marks the loudest background event in 14.8 days.

Table 5. Results for the 44 confident O3a events of GWTC-2.1, in GPS order. $\mathcal{M}_{\text{det}}$ is the detector-frame chirp mass. The catalogue FAR is in yr^{-1} . FAR here is the upper limit or estimate from the 14.8-day background, in yr^{-1} .

EVENT	$\mathcal{M}_{\text{DET}}$	IN TRAINING RANGE	CATALOGUE SNR	CATALOGUE FAR	STATISTIC	FAR HERE	RESULT
GW190403_051519	74.1	no	7.6	7.7	5.1	1.16e+03	47 louder
GW190408_181802	23.9	yes	14.6	1.0e-05	38.2	< 25	louder than all background
GW190412	15.3	yes	19.8	1.0e-05	42.2	< 25	louder than all background
GW190413_052954	38.2	no	9.0	0.82	11.4	< 25	louder than all background
GW190413_134308	53.9	no	10.6	0.18	10.0	< 25	louder than all background
GW190421_213856	45.5	no	10.7	2.8e-03	13.6	< 25	louder than all background
GW190425	1.5	no	12.4	0.034	—	—	not analysed (one detector)
GW190426_190642	131.5	no	8.7	4.1	2.5	1.68e+04	678 louder
GW190503_185404	37.8	no	12.2	1.0e-05	19.6	< 25	louder than all background
GW190512_180714	18.7	yes	12.7	1.0e-05	25.9	< 25	louder than all background
GW190513_205428	30.5	no	12.5	1.3e-05	28.4	< 25	louder than all background
GW190514_065416	47.7	no	8.0	2.8	5.6	766	31 louder
GW190517_055101	35.2	no	10.8	3.5e-04	16.4	< 25	louder than all background
GW190519_153544	64.2	no	15.9	1.0e-05	18.4	< 25	louder than all background
GW190521	98.7	no	14.3	1.3e-03	5.3	939	38 louder
GW190521_074359	39.7	no	25.9	1.0e-05	44.8	< 25	louder than all background
GW190527_092055	34.4	no	8.0	0.23	10.6	< 25	louder than all background
GW190602_175927	71.5	no	13.2	1.0e-05	9.6	< 25	louder than all background
GW190620_030421	57.2	no	12.1	0.011	—	—	not analysed (one detector)
GW190630_185205	29.6	yes	16.4	1.0e-05	—	—	not analysed (one detector)
GW190701_203306	55.5	no	11.2	5.7e-03	10.6	< 25	louder than all background
GW190706_222641	73.0	no	13.4	5.0e-05	15.4	< 25	louder than all background
GW190707_093326	9.8	yes	13.1	1.0e-05	18.2	< 25	louder than all background
GW190708_232457	15.6	yes	13.4	3.1e-04	—	—	not analysed (one detector)
GW190719_215514	36.7	no	7.9	0.63	11.6	< 25	louder than all background
GW190720_000836	10.4	yes	10.9	1.0e-05	4.3	2.47e+03	100 louder
GW190725_174728	8.9	yes	9.1	0.46	1.2	6.59e+04	2668 louder
GW190727_060333	44.7	no	11.7	1.0e-05	21.5	< 25	louder than all background
GW190728_064510	10.1	yes	13.1	1.0e-05	23.1	< 25	louder than all background
GW190731_140936	46.3	no	8.8	0.33	8.9	24.7	1 louder
GW190803_022701	42.5	no	9.3	0.073	10.4	< 25	louder than all background
GW190805_211137	61.2	no	8.1	0.63	4.3	2.47e+03	100 louder
GW190814	6.4	no	25.3	1.0e-05	33.8	< 25	louder than all background
GW190828_063405	33.9	yes	16.5	1.0e-05	39.8	< 25	louder than all background
GW190828_065509	17.3	yes	10.2	3.5e-05	12.2	< 25	louder than all background
GW190910_112807	43.2	no	14.5	2.9e-03	—	—	not analysed (one detector)
GW190915_235702	32.2	yes	13.1	1.0e-05	24.5	< 25	louder than all background
GW190916_200658	47.6	no	8.1	4.7	10.5	< 25	louder than all background
GW190917_114630	4.3	no	8.3	0.66	—	—	no event
GW190924_021846	6.4	no	12.0	1.0e-05	0.1	2.15e+05	8686 louder
GW190925_232845	18.6	yes	9.7	7.2e-03	—	—	not analysed (one detector)
GW190926_050336	37.8	no	8.1	1.1	10.1	< 25	louder than all background
GW190929_012149	54.5	no	9.7	0.16	5.7	642	26 louder
GW190930_133541	9.9	yes	9.7	0.012	3.9	3.88e+03	157 louder

These values are upper limits and demonstrations, not event significances. The background covers only April–June 2019, while the events extend to September 2019. Each background excerpt received a single random time slide. Data vetoed by data-quality criteria, and the ± 10 s around known events, are absent from the noise file.

7 Discussion

7.1 Comparison with published searches

On the real-noise dataset, the search is statistically indistinguishable from the published PyCBC result at one false alarm per month, exceeds it by 4–8% at higher false-alarm rates, and exceeds the best machine-learning submission to the challenge by 40% at one false alarm per month. A comparable level has been reported since the challenge by AResGW [Nousi et al. 2023]. The comparison with PyCBC carries two qualifications. First, the published PyCBC run used an aligned-spin template bank for a population with precession and higher-order modes. Second, as a production search it includes signal-consistency tests that cost sensitivity in Gaussian noise. It is therefore a reference search rather than an optimal filter, and the present results should not be read as evidence that neural networks outperform matched filtering in

general. Section 5.7 establishes, conversely, that the published curves are not depressed by event-timing errors, so the comparison is not flattered by a handicap of the other submissions.

7.2 Standing relative to the challenge submissions

MLGWSC-1 did not define a single ranking. The challenge paper compares submissions by their sensitive distance at 1, 10 and 100 false alarms per month and, as a secondary criterion, by the wall-clock time required to process a month of data. Table 6 places the present search among the six submissions on this basis.

Table 6. Sensitive distance in Mpc, with the rank among all seven searches in parentheses, at 10 and 1 false alarms per month, and the time to process one month of data (one foreground or background set). Published values are taken from the challenge’s released evaluation files; runtimes are those reported in the challenge paper, measured on RTX 2070 Super GPUs (machine-learning searches) or normalised to 16 CPU cores (PyCBC).

SEARCH	TYPE	DATASET 1, 10 / MONTH	DATASET 1, 1 / MONTH	DATASET 4, 10 / MONTH	DATASET 4, 1 / MONTH	RUNTIME PER MONTH
This work (calibrated, $\Delta t = 0.2$ s)	ML	2,772 (1)	2,632 (1)	1,668 (1)	1,532 (2)	≈ 1.1 h network evaluation on an RTX 3500 Ada laptop GPU, excluding whitening; not measured on the challenge hardware
PyCBC	Matched filter	2,551 (2)	2,492 (2)	1,610 (2)	1,544 (1)	≈ 43 h on 16 CPU cores (dataset 4); comparable to Virgo-AUTH on dataset 1 (smaller bank)
TPI FSU Jena	ML	2,472 (3)	2,363 (3)	— not reached	—	≈ 1 h on one GPU
Virgo-AUTH	ML	2,318 (4)	2,116 (4)	1,400 (3)	1,092 (4)	≈ 1.5 h on one GPU
cWB	Unmodelled	— not submitted	—	1,332 (4)	1,229 (3)	Not directly comparable (CPU core-seconds, foreground only)
CNN-Coinc	ML	1,045 (5)	1,002 (5)	621 (6)	— not reached	≈ 4 h on one GPU
MFCNN	ML	852 (6)	779 (6)	999 (5)	650 (5)	≈ 11.5 h on eight GPUs

Were it entered, the search would rank first on dataset 1 at all three false-alarm rates. On dataset 4 it would rank first at 100 and 10 false alarms per month; at one per month it would rank second, statistically indistinguishable from PyCBC. Among the machine-learning submissions it would rank first on both datasets at every false-alarm rate. On dataset 4 its margin over the leading machine-learning submission, Virgo-AUTH, amounts to 13%, 19% and 40% at 100, 10 and 1 false alarms per month; the challenge paper had reported that the best machine-learning search reached 70% of the PyCBC sensitive distance on this dataset. The ordering among machine-learning submissions is unchanged if the original timing estimator and the 0.3 s tolerance used by Virgo-AUTH are adopted instead (Table 2).

This placement is subject to four qualifications:

- **Retrospective design:** challenge participants submitted without knowledge of the test data, of the evaluation outcomes, or of the literature that followed. The present search was designed with the benefit of all three. The pre-registered blind confirmations (Section 5.3) guard against adaptation to the particular test months, but not against this informational advantage.
- **Incomplete coverage:** datasets 2 and 3 were not analysed, and no statement is made about them.
- **Post-hoc timing:** the calibrated timing estimator was introduced after test-set inspection (Section 8), although both blind confirmations support it.
- **Runtime:** the runtime was not measured on the challenge hardware. On the RTX 2070 Super GPUs used by the organisers we would expect roughly 2–3 h per month (not measured), which would place the search between Virgo-AUTH and CNN-Coinc in computational cost.

Since the challenge, AResGW [Nousi et al. 2023] has reported sensitivity exceeding the published PyCBC curve on dataset 4 at all false-alarm rates, subject to the same qualification concerning PyCBC’s aligned-spin bank. Because its results are presented only graphically, a quantitative comparison is not possible. The present search should therefore be regarded as reaching a level similar to that of this later work, rather than exceeding it.

7.3 Interpretation of the mass dependence

The efficiency analysis suggests a simple explanation for both the advantage and the shortfall. For heavy systems, the signal in band lasts less than a second, so the input window captures essentially all of it. The network can then exploit the two-detector data jointly, whereas the reference pipeline first forms single-detector triggers. For light systems, a large part of the signal-to-noise ratio accumulates during an inspiral that lasts several seconds before merger. A matched filter integrates over that whole inspiral; a one-second window cannot. Longer or multi-resolution inputs would therefore be the most direct way to improve sensitivity to light systems.

7.4 Computational cost

The network requires about 2 GFLOP per window, and the search evaluates ten windows per second of data. On the laptop GPU used here (NVIDIA RTX 3500 Ada), throughput was about 2,000 windows per second when the GPU was shared with training, about 3,100 under the manufacturer’s quiet thermal profile, and about 6,500 when it was dedicated at full power. Scoring a month of foreground and background therefore took between about 2 and 10 hours. The firmware thermal profile alone changed

throughput by a factor of about two. Training took 3.3 hours (dataset 1) and 5.4 hours (dataset 4, partly on a shared GPU). A cheaper first-stage screen and compiled half-precision inference would be required for operational use.

7.5 Limitations

- **Test-set exposure:** the calibrated timing estimator was prompted by test-set diagnostics. Both blind confirmations support it, but the calibrated test-set numbers themselves are not blind.
- **Duration:** results are restricted to one month per dataset. False-alarm rates below one per month are not resolved, and the threshold at one per month rests on very few background events.
- **Scope:** datasets 2 and 3 of the challenge were not analysed.
- **Numerical precision:** the statistic is computed in bfloat16, which quantises it in steps of 0.06–0.13 at high values. Ties are counted as not found, which is conservative.
- **Real events:** the results of Section 6 are upper limits obtained with a short, epoch-mismatched background.
- **Review:** the analysis has been reviewed internally (Section 4.4) but not by external domain experts.

8 Chronology of analysis decisions and test-set exposure

Table 7 lists, in order, every decision that might have been informed by test data. Highlighted rows mark exposures to test-set results.

Table 7. Chronology (local time).

TIME	EVENT	CONSEQUENCE FOR THE ANALYSIS
27 Sep 23:58	Dataset-1 training started	Configuration fixed before any test data existed
28 Sep 00:56	Pipeline test on one chunk of the dataset-1 test set with a mid-training checkpoint	Software test only; no parameter was changed
03:23–03:45	Dataset-4 training without glitch oversampling, stopped after a validation collapse	Decision based on validation data only
03:45–09:06	Dataset-4 training with glitch-centred negatives	Final dataset-4 model
10:29, ~11:40	Real-event statistic (Section 6) changed from a window-level to a cluster-level definition after the first real-event results	Twenty-six events above background under both definitions
~11:15	Diagnostic of dataset-1 test foreground events revealed mis-timed true signals	Prompted the calibrated estimator
11:34–11:50	Timing estimator selected and calibrated on validation data (both datasets)	Offsets and $\delta = 4$ fixed without test data
12:05	Dataset-1 blind configuration registered	Blind data generated afterwards
14:32, 15:20	Official test results computed (original timing, as launched), then re-clustered with the calibrated estimator	Both versions reported
15:56	Dataset-1 blind result	Within 1% of the calibrated test result
16:37	Dataset-4 blind configuration and success criterion registered	Blind data generated afterwards
16:38–19:38	First blind execution stopped after two of three parts (19:35); deterministic rerun with checkpoints started at 19:38	No configuration change; no interim result inspected
29 Sep 00:31	Dataset-4 blind result	Criterion met (+2.9%)

9 Reproducibility

All code, configuration and pre-registration files are kept in the project repository; `src/run_pipeline.sh` lists every stage in order with the arguments used. The environment is the conda environment `gw`: Python 3.11, gwpy 4.0.2, pycbc 2.11.0, lalsuite and PyTorch 2.14 with CUDA 12.6. Every random seed is fixed, and the official test sets are regenerated from the published seeds. The principal steps for dataset 4 are:

```
python src/make_noise_ds4.py --test-end 1244600712 --train-days 5 --val-days 1 --out data/ds4_noise
python src/make_signals.py --dataset 4 --n 60000 --seed 54321 --psd-library data/ds4_noise/psd_library.npz --out data/signals
python src/train.py --train-noise "data/ds4_noise/train/noise_*.npz" --val-noise "data/ds4_noise/val/noise_*.npz" \
--signals data/signals_ds4.npz --out results/ds4_hard --steps 40000 --hard-frac 0.4
python src/calibrate_timing.py --model results/ds4_hard/best.pt --val-noise "data/ds4_noise/val/noise_*.npz" \
--signals data/signals_ds4.npz --out results/ds4_hard/timing.json --n 800
python src/prepare_test_ds4.py --seed 2514409456 --out data/test_ds4
python src/run_eval.py --model results/ds4_hard/best.pt --test data/test_ds4 --dataset 4 --out results/ds4_hard/eval_calibrated
--timing results/ds4_hard/timing.json
python src/run_eval.py --model results/ds4_hard/best.pt --test data/test_ds4 --dataset 4 --out results/ds4_hard/eval_calibrated
--windows-dir results/ds4_hard/eval_calibrated --timing results/ds4_hard/timing.json --var 0.2
python src/recheck.py --run results/ds4_hard/eval_calibrated --test data/test_ds4
python src/blind_ds4.py --prereg results/blind_check_ds4_preregistration.json --out data/blind_ds4 --results results/ds4_hard
```

Dataset 1 uses `make_noise.py`, `make_signals.py` and `prepare_test.py`. The registered checksums are MD5 6ef639de... (dataset-1 model), f98e2707... (dataset-4 model) and 749e9293... (dataset-4 timing file). Figures are produced by `report_figures.py` and `efficiency.py`.

Appendix A: Internal review findings and responses

SEVERITY	FINDING	RESPONSE
Major	Test-set results influenced the method: the timing estimator, the real-event statistic, and a pipeline test during training	Disclosed (Section 8); original-timing results reported alongside; two pre-registered blind confirmations
Major	False-alarm rates of the real events are weak upper limits	Reported as upper limits, with the epoch, time-slide and veto caveats stated
Minor	bfloat16 quantisation of the statistic	Disclosed; ties counted as not found (conservative)
Minor	Short final whitening chunk at segment ends could produce outliers	Checked: none of the 10 loudest background events in either month lies within 1,600 s of a segment end
Minor	Misleading log message on the equality of injection files	Equality check now hashes the datasets rather than the file bytes
Minor	Generator error output discarded; cached statistics not tied to a model	Error output now logged; model checksum verified before reuse
Note	Matching tolerance differs between submissions (0.2 s and 0.3 s)	Both tolerances reported
Note	No window-time or stride errors found	—

Appendix B: Hyperparameters

COMPONENT	SETTING
Whitening	512 s chunks, 8 s overlap each side; Welch PSD, 4 s segments, 50% overlap, median; zero below 18 Hz; Tukey taper within the discarded overlap
Input	$2 \times 2,048$ samples (1 s, H1 and L1); bank signals stored from 1.75 s before to 0.25 s after merger
Architecture	Stem Conv1d(2 → 32, k = 31); 5 stages $\times$ 2 residual blocks, widths 32/64/128/256/256, kernel 7, stride 2 from stage 2; mean @ max pooling; head 512 → 64 → 1; 4.12 M parameters
Optimisation	AdamW, peak learning rate 10^{-3} , weight decay 10^{-4} , one-cycle schedule with 5% warm-up; 40,000 steps; batch 512 (half with signal); bfloat16 autocast
Signals	Merger position uniform in [0.45, 0.95] s; SNR uniform on [10, 25] annealed to [5, 20] over 40% of training
Glitch oversampling (dataset 4)	40% of noise-only windows glitch-centred; glitch = whitened sample above 6σ in a 1,024-sample block; glitch offset uniform in the central 75% of the window
Validation and selection	200,000 noise windows and 5,000 held-out signals at SNR 6/8/10; checkpoint maximising detection at SNR 8 and a per-window false-positive probability of 10^{-4} , evaluated every 2,000 steps
Search	Stride 205 samples; trigger threshold = 99.8th percentile of background window statistics; clustering gap 1 s; plateau δ = 4; offsets +0.066 s (dataset 1) and +0.086 s (dataset 4); Δt = 0.2 s (primary) and 0.3 s

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MLGWSC-1 code and released results: github.com/gwastro/ml-mock-data-challenge-1; doi:10.5281/zenodo.7107410

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